

Multi-species Catalogues of Regulatory Elements: ENCODE

Session 2

T6: Making the most of cis-regulatory annotations for genome-scale analysis

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Session map

1	The ENCODE Project	• General description • Available datasets
1	The ENCODE Data Portal	• Navigation • Accessing experimental data and metadata • Defining and accessing reference epigenome data
1	Hands-on	
2	Registry of cCREs	• cCREs definition and identification • Classification • Biosample classification
2	The SCREEN browser	• Accessing cCREs • Accessing experimental data and metadata
2	Hands-on	

Coffee break



1

ENCODE: Experimental data



Have you used ENCODE data?



The ENCODE project

- ✓ The Encyclopedia of DNA Elements (ENCODE) Consortium is an ongoing international collaboration of research groups launched on 2003.
- ✓ It aims to build a comprehensive parts list of **functional elements** in the **human** and **mouse genomes**.
- ✓ It maps genomic activity across tissues, cell types, developmental stages, and experimental conditions.
- ✓ ENCODE data help interpret non-coding regions by connecting genomic coordinates to regulatory evidence such as chromatin accessibility, histone marks, transcription factor binding, RNA expression, methylation, and 3D chromatin structure.
- ✓ By the end of 2024, it includes results from 23,000+ functional genomics experiments, 800+ functional elements characterization experiments and 60,000+ results from integrative computational analyses, available on an open-access data-portal.

Standardised data analysis pipelines

Standardised metadata

The ENCODE project and the ENCODE portal

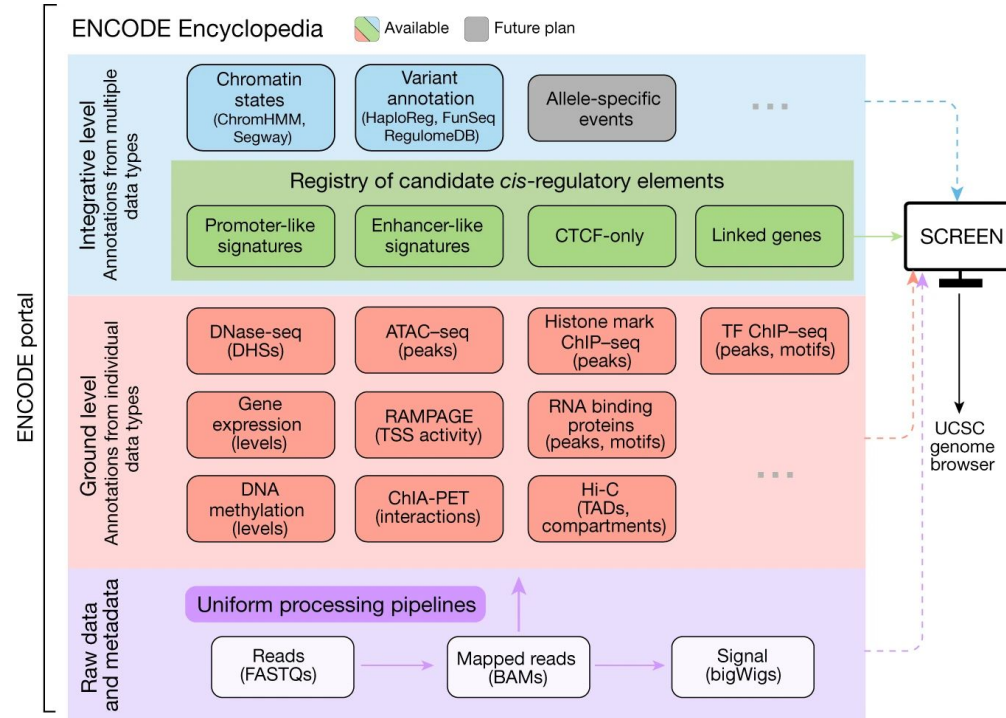


Figure extracted from *Expanded encyclopaedias of DNA elements in the human and mouse genomes*¹

¹The ENCODE Project Consortium., Moore, J.E., Purcaro, M.J. et al. Expanded encyclopaedias of DNA elements in the human and mouse genomes. Nature 583, 699–710 (2020). <https://doi.org/10.1038/s41586-020-2493-4>

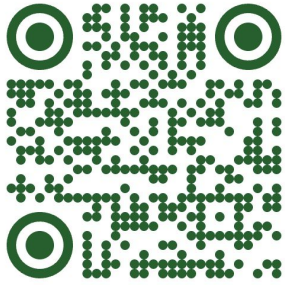
The ENCODE project and the ENCODE portal



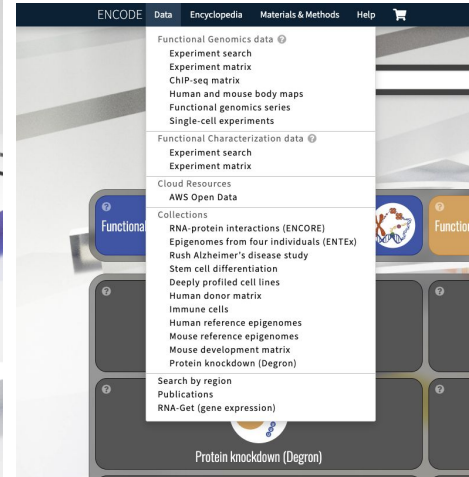
Figure extracted from *Data navigation on the ENCODE portal*²

²Kagda, M.S., Lam, B., Litton, C. et al. Data navigation on the ENCODE portal. Nat Commun 16, 9592 (2025). <https://doi.org/10.1038/s41467-025-64343-9>

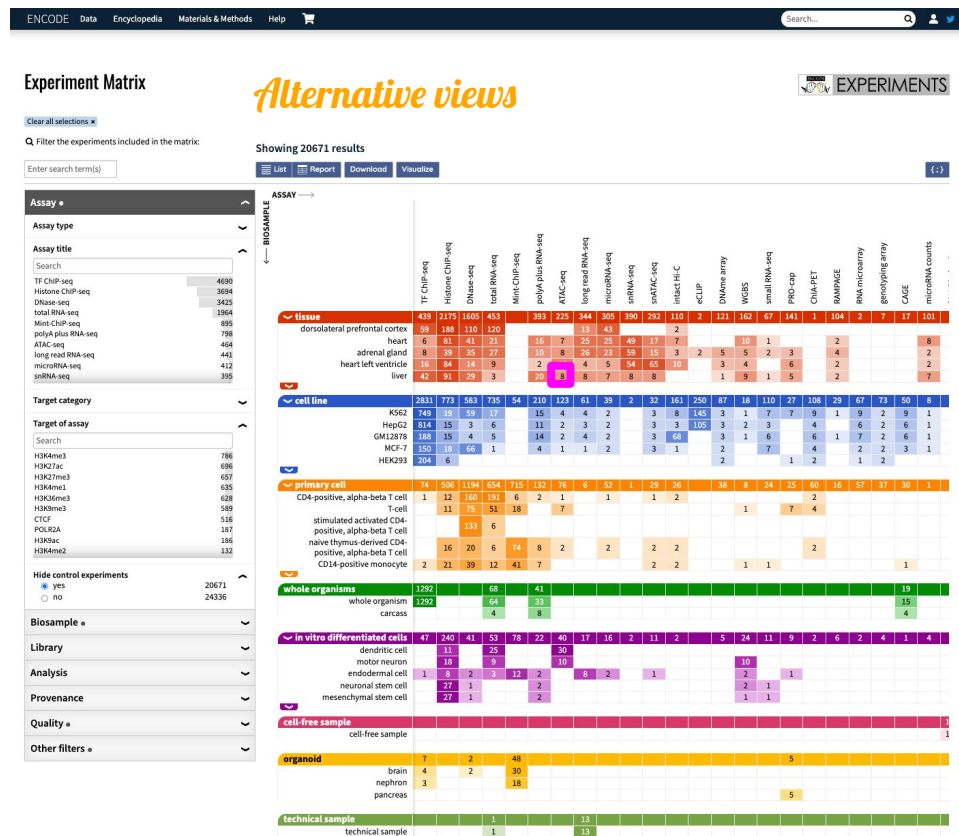
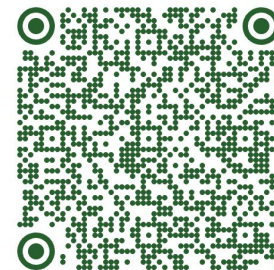
The ENCODE project and the ENCODE portal



<https://www.encodeproject.org/>



The ENCODE project and the ENCODE portal



Filtering sidebar

Available experiments

The ENCODE project and the ENCODE portal



ENCODE Data Encyclopedia Materials & Methods Help

Search...

Experiment search

Showing 7 of 7 results

Download JSON

Report Experiment matrix Download Visualize

Number of displayed results: 25 50 100 200

Assay type

Assay title

Search

Histone ChIP-seq 66

DNase-seq 27

polyA plus RNA-seq 17

WGBS 8

ATAC-seq 7

TF ChIP-seq 7

microRNA counts 7

microRNA-seq 7

long read RNA-seq 4

total RNA-seq 1

ATAC-seq x

Hide control experiments

Biosample

Organism

Mus musculus 7

Homo sapiens 1

Mus musculus x

Biosample classification

Biosample

Search

forebrain 7

heart 7

hindbrain 7

liver 7

midbrain 7

dendritic cell 6

embryonic facial prominence 5

limb 5

neural tube 5

intestine 4

liver x

Organ

Systems

Sex

ATAC-seq in liver

Mus musculus strain B6NTac;B6NCR/Lap liver tissue postnatal (0 days)

Lab: Bing Ren, UCSD

Project: ENCODE

Reference Epigenome: ENCSR687SNT

Organism Development Series: ENCSR013PMI

ATAC-seq in liver

Mus musculus strain B6NCR/liver tissue embryo (16.5 days)

Lab: Bing Ren, UCSD

Project: ENCODE

Reference Epigenome: ENCSR826FLO

Organism Development Series: ENCSR326DKM

ATAC-seq in liver

Mus musculus strain B6NCR/liver tissue embryo (15.5 days)

Lab: Bing Ren, UCSD

Project: ENCODE

Reference Epigenome: ENCSR871WIE

Organism Development Series: ENCSR326DKM

ATAC-seq in liver

Mus musculus strain B6NTac;B6NCR/Lap liver tissue embryo (14.5 days)

Lab: Bing Ren, UCSD

Project: ENCODE

Reference Epigenome: ENCSR248SXA

Organism Development Series: ENCSR326DKM

ATAC-seq in liver

Mus musculus strain B6NCR/liver tissue embryo (13.5 days)

Lab: Bing Ren, UCSD

Project: ENCODE

Reference Epigenome: ENCSR266TXT

Organism Development Series: ENCSR326DKM

ATAC-seq in liver

Mus musculus strain B6NCR/liver tissue embryo (12.5 days)

Lab: Bing Ren, UCSD

Project: ENCODE

Reference Epigenome: ENCSR192PSV

Organism Development Series: ENCSR326DKM

Add all items to cart

Experiment ENCSR609OHJ released 2

Experiment ENCSR255XTC released 4

Experiment ENCSR465PYP released 1 3

Experiment ENCSR032HKE released 1 3

Experiment ENCSR343TXK released 4

Experiment ENCSR302LV released 3

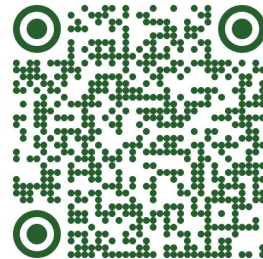
ATAC-seq from
mouse liver
filters applied

Experiment

Status

Audits

The ENCODE project and the ENCODE portal



Metadata

Replicates
summary

ENCODEDataEncyclopediaMaterials & MethodsHelp

Search...

Experiments / ATAC-seq / Mus musculus / liver

Experiment summary for ENCSR465PYP

doi:10.17989/ENCSR465PYP

13

Summary

Status: released

Assay: ATAC-seq

Biosample summary: Mus musculus strain B6NCrI liver tissue embryo (15.5 days)

Biosample Type: tissue

Replication type: isogenic

Description: ATAC-seq on embryonic e15.5 mouse liver

Nucleic acid type: DNA

Size range: 200-750

Lysis method: ATAC buffer

Fragmentation methods: chemical (Nextera tagmentation)

Size selection method: SPRI beads

Platform: Illumina HiSeq 4000

Attribution

Lab: Bing Ren, UCSD

Award: U54HG006997 (Bing Ren, UCSD)

Project: ENCODE

External resources: GEO:GSE173061

Aliases: bing-ren:e15.5_liver_ATAC-seq

Date submitted: March 14, 2017

Date released: October 10, 2017

Reference Epigenome: ENCSR871WIE

Organism Development Series: ENCSR326DKM

Annotation (gkmsvm-model): ENCSR085DVU, ENCSR132BUH

Tags: ENCODE ENCyclopedia

Isogenic replicates

Isogenic replicate	Technical replicate	Summary	Biosample	Library
1	1	Mus musculus strain B6NCrI mixed sex embryo (15.5 days) liver tissue	ENCB5734ENZ	ENCLB423CNX
2	1	Mus musculus strain B6NCrI mixed sex embryo (15.5 days) liver tissue	ENCB5966PAM	ENCLB633ZWN

Files

Genome browserAssociation graphFile details

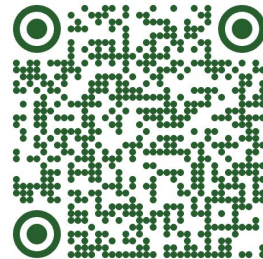
Filter files

File format

mm10UCSCVisualizeDownload

Displaying 34 of 34 files

The ENCODE project and the ENCODE portal



Files

Genome browser

Association graph

File details

mm10

UCSC

Visualize

Download

Displaying 34 of 34 files

Raw sequencing data (4 Files)

Isogenic replicate	Library	Accession	File type	Run type	Read	Output type	Lab	Date added	File size	File status
1	ENCLB423CNX	ENCFF329VCX	fastq	PE50nt	1	reads	Bing Ren, UCSD	2017-03-14	1.41 GB	released
		ENCFF290ZBP	fastq	PE50nt	2	reads	Bing Ren, UCSD	2017-03-14	1.53 GB	released
2	ENCLB633ZWN	ENCFF341HRL	fastq	PE50nt	1	reads	Bing Ren, UCSD	2017-03-14	888 MB	released
		ENCFF489XAT	fastq	PE50nt	2	reads	Bing Ren, UCSD	2017-03-14	968 MB	released

ENCODE4 v1.8.0 mm10 (ENCAN375YTA) processed data (30 Files)

released

1 3

Accession	Default	File type	Output type	Isogenic replicate	Mapped read length	Genome assembly	Date added	File size	File status
ENCFF0620ME	★	bed narrowPeak	pseudoreplicated peaks	1, 2		mm10	2020-08-18	2.06 MB	released
ENCFF82ZVOQ	★	bigBed narrowPeak	pseudoreplicated peaks	1, 2		mm10	2020-08-18	4.65 MB	released
ENCFF479CWC	★	bigWig	signal p-value	1, 2		mm10	2020-08-18	599 MB	released
ENCFF063ZRU		bigBed narrowPeak	replicated peaks	1, 2		mm10	2020-08-18	4.54 MB	released
ENCFF565NOC		bam	alignments	1	51	mm10	2020-08-18	2.23 GB	released
ENCFF827GVE		bed narrowPeak	pseudoreplicated peaks	1		mm10	2020-08-18	1.72 MB	released
ENCFF809PRS		bigWig	signal p-value	1		mm10	2020-08-18	364 MB	released
ENCFF419PZT		bed idr_ranked_peak	IDR ranked peaks	1, 2		mm10	2020-08-18	4.32 MB	released
ENCFF152WBR		bed idr_ranked_peak	IDR ranked peaks	1, 2		mm10	2020-08-18	4.28 MB	released
ENCFF197LVY		bigWig	signal p-value	2		mm10	2020-08-18	225 MB	released
ENCFF507ZEN		bed narrowPeak	IDR thresholded peaks	1		mm10	2020-08-18	991 kB	released
ENCFF385MYQ		bigBed narrowPeak	IDR thresholded peaks	1		mm10	2020-08-18	2.15 MB	released
ENCFF232WSN		bed narrowPeak	replicated peaks	1, 2		mm10	2020-08-18	2.02 MB	released
ENCFF717HRV		bam	unfiltered alignments	1	51	mm10	2020-08-18	5.31 GB	released

Filter files

File format

bigBed narrowPeak 8

bed narrowPeak 8

bigWig 6

fastq 4

bam 4

bed idr_ranked_peak 4

Output type

pseudoreplicated peaks 6

IDR thresholded peaks 6

reads 4

IDR ranked peaks 4

signal p-value 3

fold change over control 3

replicated peaks 2

alignments 2

unfiltered alignments 2

conservative IDR thresholded peaks 2

Replicates

1, 2 12

1 11

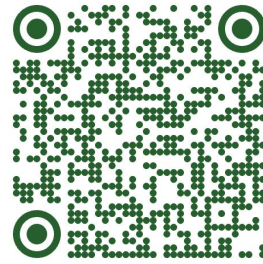
2 11

Raw files

Processed files

EMBL-EBI

The ENCODE project and the ENCODE portal



ENCODE Data Encyclopedia Materials & Methods Help

Search...

Experiments / [ATAC-seq](#) / [Mus musculus](#) / [liver](#)

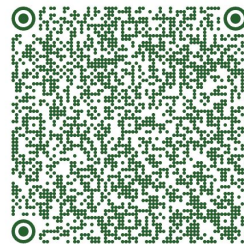
Experiment summary for ENCSR465PYP

doi:10.17989/ENCSR465PYP

Summary	Attribution
Status: ● released	Lab: Bing Ren, UCSD
Assay: ATAC-seq	Award: U54HG006997 (Bing Ren, UCSD)
Biosample summary: <i>Mus musculus</i> strain B6Ncrl liver tissue embryo (15.5 days)	Project: ENCODE
Biosample Type: tissue	External resources: GEO:GSE173061
Replication type: isogenic	Aliases: bing-ren:e15.5_liver_ATAC-seq
Description: ATAC-seq on embryonic e15.5 mouse liver	Date submitted: March 14, 2017
Nucleic acid type: DNA	Date released: October 10, 2017
Size range: 200-750	Reference Epigenome: ENCSR871WIE
Lysis method: ATAC buffer	Organism Development Series: ENCSR326DKM
Fragmentation methods: chemical (Nextera tagmentation)	Annotation (gkmsvm-model): ENCSR085DVU , ENCSR132BUH
Size selection method: SPRI beads	Tags:
Platform: Illumina HiSeq 4000	

Reference epigenomes: grouped data from a particular tissue, cell line, primary cell, in vitro differentiated cell biosamples organized following guidelines set out by IHEC (<https://ihec-epigenomes.org/>)

The ENCODE Reference Epigenomes



Reference

List

Showing 31

Home

Clear all

Filter

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Filter

Filter

Filter

Functional Genomics data

- Experiment search
- Experiment matrix
- ChIP-seq matrix
- Human and mouse body maps
- Functional genomics series
- Single-cell experiments

Functional Characterization data

- Experiment search
- Experiment matrix

Cloud Resources

- AWS Open Data

Collections

- RNA-protein interactions (ENCORE)
- Epigenomes from four individuals (ENTEX)
- Rush Alzheimer's disease study
- Stem cell differentiation
- Deeply profiled cell lines
- Human donor matrix
- Immune cells
- Human reference epigenomes
- Mouse reference epigenomes**
- Mouse development matrix
- Protein knockdown (Degron)

Search by region

- Publications
- RNA-Get (gene expression)

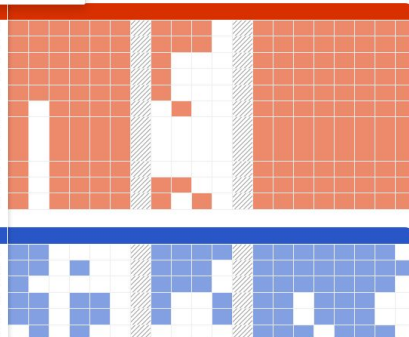
ATAC-seq
WGBS
TF ChIP-seq
CTCF
POLR2A
EP300
POLR2AphosphoS5
Histone ChIP-seq
H3K4me1
H3K4me3
H3K27ac
H3K27me3
H3K36me3
H3K9me3
H3K9ac
H3K4me2

tissue

- heart
- liver
- forebrain
- midbrain
- hindbrain
- limb
- embryonic
- facial
- prominence
- neural tube
- kidney
- lung

cell line

- MEL
- CH12.LX
- ES-Bruce4
- G1E
- G1E-ER4
- ES-E14



Project data from mouse tissue, cell line, primary cell, and in vitro differentiated cell biosamples organized as reference epigenomes following guidelines set out by IHEC.

Homo sapiens

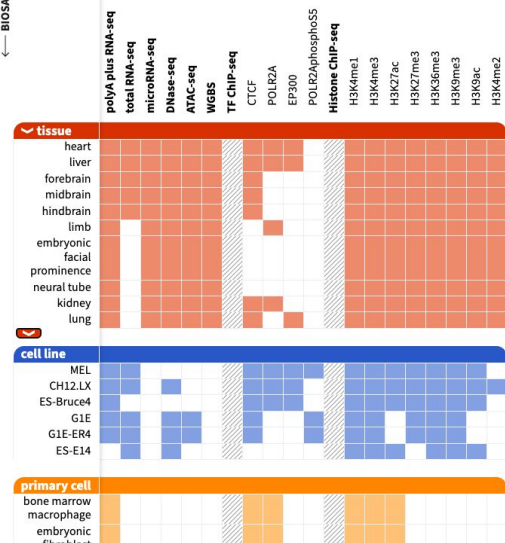
Mus musculus

Clear all body map selections



ASSAY →

BIOSAMPLE ←



The ENCODE Reference Epigenomes



Experiment search

Clear all selections x

Assay ▾

Assay type ▾

Assay title

Search

Histone ChIP-seq 62

polyA plus RNA-seq 8

ATAC-seq 7

WGBS 7

microRNA-seq 7

TF ChIP-seq 6

DNase-seq 4

total RNA-seq 1

Target category ▾

Target of assay

Search

H3K27ac 8

H3K27me3 8

H3K36me3 8

H3K4me1 8

H3K4me3 8

H3K9ac 8

H3K4me2 7

H3K9me3 7

CTCF 3

EP300 2

Hide control experiments ▾

Biosample ▾

Library ▾

Analysis ▾

Provenance ▾

Quality ▾

Other filters ▾

Showing 25 of 102 results

Report Experiment matrix Download Visualize

Number of displayed results: 25 50 100 200

Add all items to cart

microRNA-seq in liver

Mus musculus strain B6NCRl liver tissue embryo (12.5 days)

Lab: Ali Mortazavi, UCI

Project: ENCODE

Reference Epigenome: [ENCSR192PSV](#)

Organism Development Series: [ENCSR103VDG](#)

Experiment

ENCSR453ESQ

released

1

ATAC-seq in liver

Mus musculus strain B6NTac;B6NCRl/Lap liver tissue postnatal (0 days)

Lab: Bing Ren, UCSD

Project: ENCODE

Reference Epigenome: [ENCSR687SNT](#)

Organism Development Series: [ENCSR013PMI](#)

Experiment

ENCSR609OHJ

released

2

ATAC-seq in liver

Mus musculus strain B6NCRl liver tissue embryo (16.5 days)

Lab: Bing Ren, UCSD

Project: ENCODE

Reference Epigenome: [ENCSR826FLO](#)

Organism Development Series: [ENCSR326DKM](#)

Experiment

ENCSR255KTC

released

4

ATAC-seq in liver

Mus musculus strain B6NCRl liver tissue embryo (15.5 days)

Lab: Bing Ren, UCSD

Project: ENCODE

Reference Epigenome: [ENCSR871WIE](#)

Organism Development Series: [ENCSR326DKM](#)

Experiment

ENCSR465PYP

released

1 3

ATAC-seq in liver

Mus musculus strain B6NTac;B6NCRl/Lap liver tissue embryo (14.5 days)

Lab: Bing Ren, UCSD

Project: ENCODE

Reference Epigenome: [ENCSR248SXA](#)

Organism Development Series: [ENCSR326DKM](#)

Experiment

ENCSR012HKE

released

1 3

ATAC-seq in liver

Mus musculus strain B6NCRl liver tissue embryo (13.5 days)

Lab: Bing Ren, UCSD

Project: ENCODE

Reference Epigenome: [ENCSR266TXX](#)

Organism Development Series: [ENCSR326DKM](#)

Experiment

ENCSR343TXK

released

4

The ENCODE Reference Epigenomes



ENCODE Data Encyclopedia Materials & Methods Help

Search...

Control experiments

1 control experiment

Add all items to cart

Accession	Control type	Biosample summary	Lab	Status	Cart
ENCSR274AQL	input library	Mus musculus strain B6NCrI liver tissue embryo (15.5 days)	Bing Ren, UCSD	released	

Files

Genome browser File details

Include deprecated files

mm10 M21 Visualize Download

Displaying 36 of 36 files

ENC SR465PYP ATAC-seq mm10 (3 Files) released 1 5

Accession	File type	Output type	Genome assembly	Date added	File size	File status
ENCF0620ME	bed narrowPeak	pseudoreplicated peaks	mm10	2020-08-18	2.06 MB	released
ENCF479CWC	bigWig	signal p-value	mm10	2020-08-18	599 MB	released
ENCF822VOQ	bigBed narrowPeak	pseudoreplicated peaks	mm10	2020-08-18	4.65 MB	released

ENC SR258RGT ChIP-seq H3K4me2 mm10 (3 Files) released 2

Accession	File type	Output type	Genome assembly	Date added	File size	File status
ENCF636RRG	bigWig	signal p-value	mm10	2020-11-08	947 MB	released
ENCF844PLJ	bed narrowPeak	pseudoreplicated peaks	mm10	2020-11-08	968 kB	released
ENCF822EZR	bigBed narrowPeak	pseudoreplicated peaks	mm10	2020-11-08	2.08 MB	released

ENC SR889MY ChIP-seq H3K9ac mm10 (3 Files) released 2

Accession	File type	Output type	Genome assembly	Date added	File size	File status
ENCF672QEI	bed narrowPeak	pseudoreplicated peaks	mm10	2020-11-07	600 kB	released
ENCF669NMA	bigBed narrowPeak	pseudoreplicated peaks	mm10	2020-11-07	1.32 MB	released
ENCF344BQF	bigWig	signal p-value	mm10	2020-11-07	988 MB	released

ENC SR577SDJ ChIP-seq H3K4me3 mm10 (3 Files) released 1

Accession	File type	Output type	Genome assembly	Date added	File size	File status
ENCF954ONV	bed narrowPeak	replicated peaks	mm10	2020-10-09	627 kB	released
ENCF414VBJ	bigWig	signal p-value	mm10	2020-10-09	856 MB	released
ENCF949VCX	bigBed narrowPeak	replicated peaks	mm10	2020-10-09	1.46 MB	released

ENC SR855NKG ChIP-seq H3K9me3 mm10 (3 Files) released

Filter files

File format

- bigWig 14
- bed narrowPeak 9
- bigBed narrowPeak 9
- tsv 2
- bed bed+ 1
- bigBed bed+ 1

Output type

- replicated peaks 10
- signal p-value 9
- pseudoreplicated peaks 8
- methylation state at CpG 2
- signal of unique reads 1
- gene quantifications 1
- plus strand methylation state at CpG 1
- minus strand methylation state at CpG 1
- microRNA quantifications 1
- plus strand signal of all reads 1
- minus strand signal of all reads 1

Replicates

- 1, 2 27
- 1 7
- 2 2

The ENCODE Reference Epigenomes

ENCODEDataEncyclopediaMaterials & MethodsHelp

liver

Filtering by: liver

Showing 25 of 47530 results

Data Type

OrganismDevelopmentSeries21

BioSampleSeries16

ReferenceEpigenome13

Project25

MultiAssayExperiment16

Audit category:

Number of displayed results:

2550100200

Add all items to cart

/documents/8001cec6-b18b-49f4-a1c5-e4579dcfc222/

Liver_STL004 excision protocol

/documents/922ef39e-4dee-4f18-b42b-4339a5592aab/

Liver_STL011 excision protocol

/documents/67beb09d-d4f3-444a-87a3-4fd51f787ad6/

Differentiation Protocol for Liver_CyT49

/documents/1ffe6905-eab2-4250-a578-0cd067875162/

Liver (primary liver from Open Chromatin ENCODE Group) data sheet for DNase experiments

/documents/4c730668-ad84-4471-9918-86011a81de07/

Mouse liver dissection and ChIP-seq preparation protocol

/documents/abb50469-5e1c-423a-b738-a6d80562517e/

BioChain Human fetal liver H71151 data sheet

/documents/47835819-06df-43b4-8556-8db0056513fc/

BioChain Human fetal liver A601605 data sheet

Reference epigenome search

Filtering by: liver

Clear all selections

Assay

Assay title

Control ChIP-seq8

Histone ChIP-seq8

polyA plus RNA-seq8

ATAC-seq7

WGBS7

Target category

Biosample

Organism

Mus musculus8

Homo sapiens5

Mus musculus

Biosample classification

tissue8

Organ

endocrine gland8

exocrine gland8

liver8

Life stage

Library

Analysis

Provenance

Quality

Other filters

Showing 8 of 8 results

Report

Number of displayed results:

2550100200

Add all items to cart

Reference epigenome series in liver (embryonic, 15.5 days)

Assays: ATAC-seq, whole-genome shotgun bisulfite sequencing, polyA plus RNA-seq, ChIP-seq, microRNA-seq

Targets: H3K4me3, H3K4me1, H3K27ac, H3K4me2, H3K36me3, H3K9ac, H3K9me3, H3K27me3

Lab: ENCODE Consortium

Project: ENCODE

Reference epigenome series in liver (embryonic, 13.5 days)

Assays: ATAC-seq, whole-genome shotgun bisulfite sequencing, polyA plus RNA-seq, ChIP-seq, microRNA-seq

Targets: H3K4me3, H3K4me1, H3K27ac, H3K4me2, H3K36me3, H3K9ac, H3K9me3, H3K27me3

Lab: ENCODE Consortium

Project: ENCODE

Reference epigenome series in liver (embryonic, 12.5 days)

Assays: ATAC-seq, whole-genome shotgun bisulfite sequencing, polyA plus RNA-seq, ChIP-seq, microRNA-seq

Targets: H3K4me3, H3K4me1, H3K27ac, H3K4me2, H3K36me3, H3K9ac, H3K9me3, H3K27me3

Lab: ENCODE Consortium

Project: ENCODE

Reference epigenome series in liver (embryonic, 14.5 days)

Assays: ATAC-seq, whole-genome shotgun bisulfite sequencing, polyA plus RNA-seq, ChIP-seq, microRNA-seq, DNase-seq

Targets: CTCF, H3K4me3, EP300, H3K4me1, H3K27ac, H3K4me2, H3K36me3, H3K9ac, H3K9me3, H3K27me3

Lab: ENCODE Consortium

Project: ENCODE

Reference epigenome series in liver (postnatal, 0 days)

Assays: ATAC-seq, whole-genome shotgun bisulfite sequencing, polyA plus RNA-seq, ChIP-seq, microRNA-seq, DNase-seq

Targets: CTCF, H3K4me3, EP300, H3K4me1, H3K27ac, H3K4me2, H3K36me3, H3K9ac, H3K27me3, H3K9me3

Lab: ENCODE Consortium

Project: ENCODE



Which of these statements are true about ENCODE?



Questions?



Hands-on





How many DNase-seq experiments from female human cerebellum tissue donors are available?





How many reference epigenomes for human hepatocytes are available?





How many ENCODE experiments are linked to forebrain E13.5?





Are there any open chromatin experiments available that are linked to forebrain E13.5?



A background image showing a dense population of cells, likely in a tissue section, with red and green fluorescence. The red fluorescence is more prominent, while the green fluorescence highlights specific cells or structures.

2

ENCODE cCREs and SCREEN

The ENCODE Registry of cCREs and SCREEN

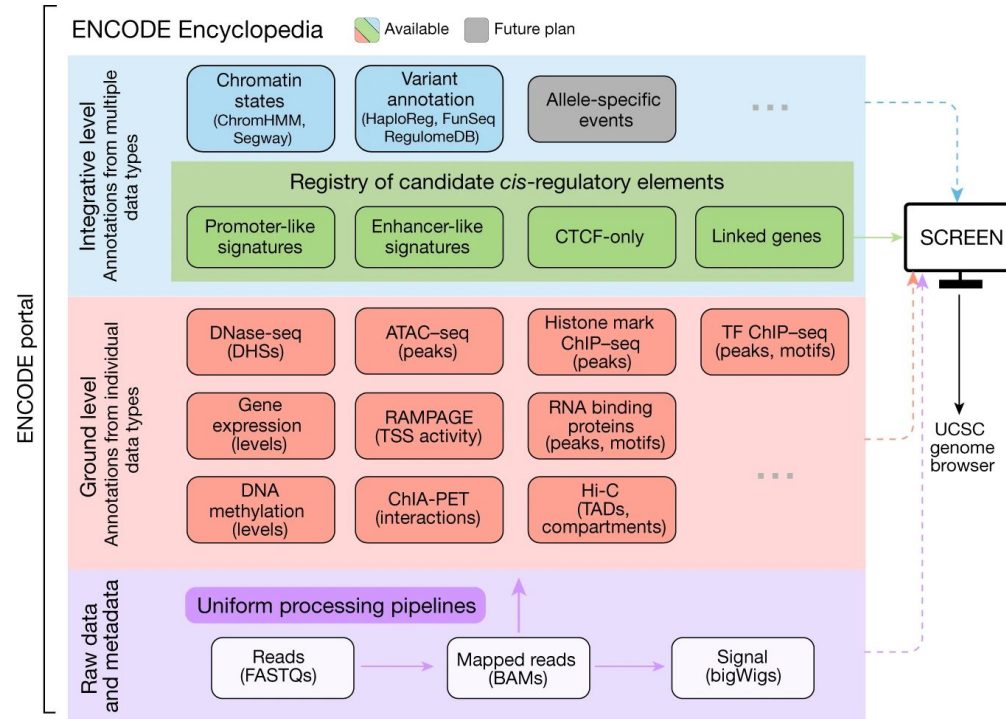


Figure extracted from *Expanded encyclopaedias of DNA elements in the human and mouse genomes*¹

¹The ENCODE Project Consortium., Moore, J.E., Purcaro, M.J. et al. Expanded encyclopaedias of DNA elements in the human and mouse genomes. Nature 583, 699–710 (2020). <https://doi.org/10.1038/s41586-020-2493-4>

The ENCODE Registry of cCREs

- ✓ A candidate cis-regulatory element is a **genomic interval** with ENCODE biochemical evidence for regulatory activity.
- ✓ The registry gives each interval a stable coordinate, cell-type agnostic class, and evidence trail.
- ✓ Context-aware: one registry element can have different activity calls across biosamples.

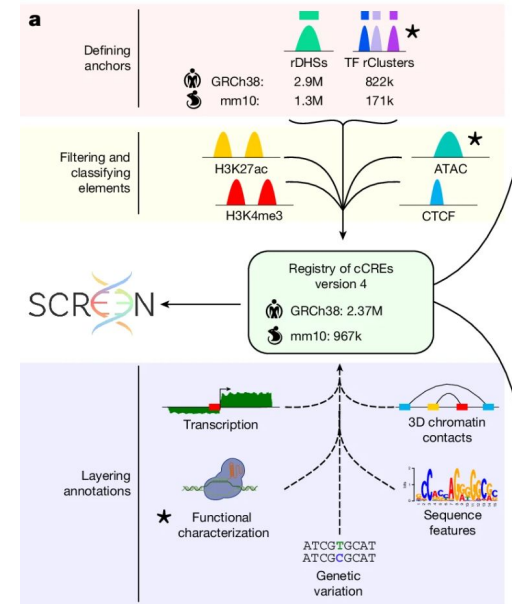


Figure extracted from *Expanded encyclopaedias of DNA elements in the human and mouse genomes*³

³Moore, J.E., Pratt, H.E., Fan, K. et al. An expanded registry of candidate cis-regulatory elements. *Nature* (2026). <https://doi.org/10.1038/s41586-025-09909-9>

The ENCODE Registry of cCREs

✓ cCREs are anchored by using complementary strategies:

- Representative DNase hypersensitivity sites (rDHSs): Cluster DNase-seq peaks across biosamples.
- Reproducible TF binding (TF rClusters): Recover protein-binding sites that may lack strong accessibility.
- Multi-mapper rDHSs: Improve coverage in duplicated and repetitive regions

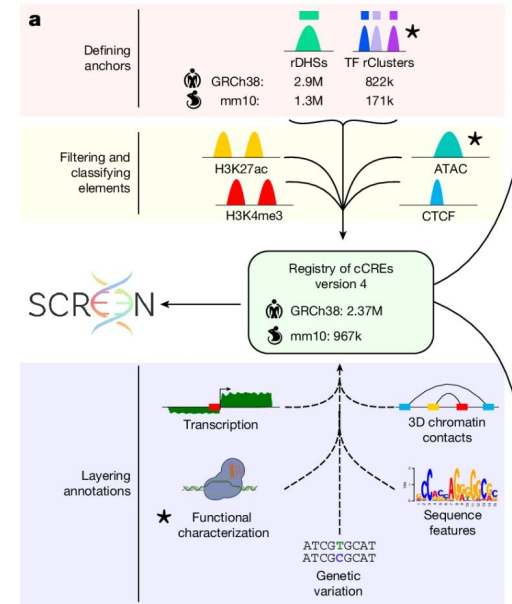


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The ENCODE Registry of cCREs

✓ cCREs are classified based on epigenomic evidence across surveyed biosamples:

- **Promoter-like cCREs:** Near a TSSs with high chromatin accessibility (CA) and H3K4me3 signal.
- **Enhancer-like cCREs:** Accessible elements with high H3K27ac signal, subdivided into TSS-proximal and TSS-distal enhancers.
- **Chromatin accessibility + H3K4me3 (CA-H3K4me3) cCREs:** Accessible elements with H3K4me3 but lacking strong H3K27ac and located away from TSSs.
- **Chromatin accessibility + CTCF (CA-CTCF) cCREs:** Accessible elements with strong CTCF binding and low histone acetylation.
- **Chromatin accessibility + transcription factor (CA-TF) cCREs:** Accessible elements overlapping transcription factor clusters but lacking strong histone modification signals.
- **Chromatin accessibility (CA) cCREs:** Accessible elements lacking strong H3K4me3, H3K27ac, or CTCF signals.
- **Transcription factor (TF) cCREs:** TF binding in the absence of detectable chromatin accessibility or histone modification signals.

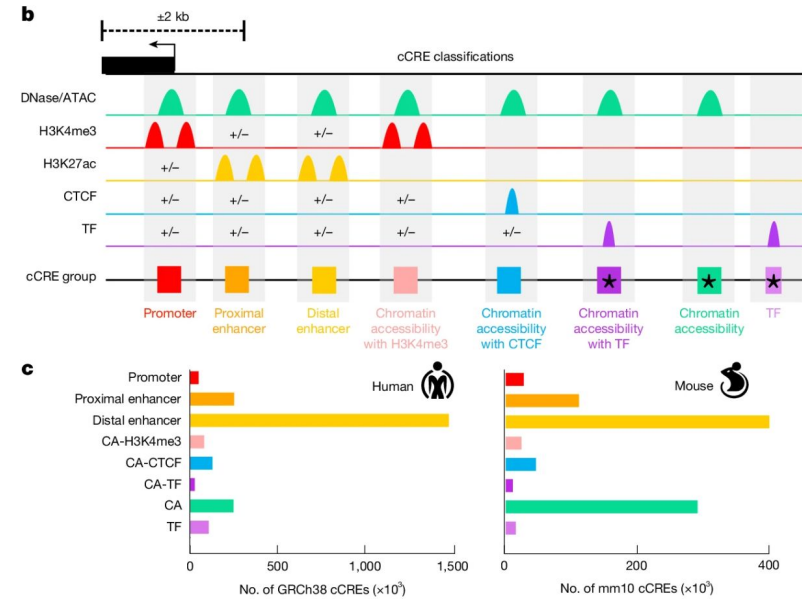


Figure extracted from *Expanded encyclopaedias of DNA elements in the human and mouse genomes*³

³Moore, J.E., Pratt, H.E., Fan, K. et al. An expanded registry of candidate cis-regulatory elements. Nature (2026). <https://doi.org/10.1038/s41586-025-09909-9>

The ENCODE Registry of cCREs

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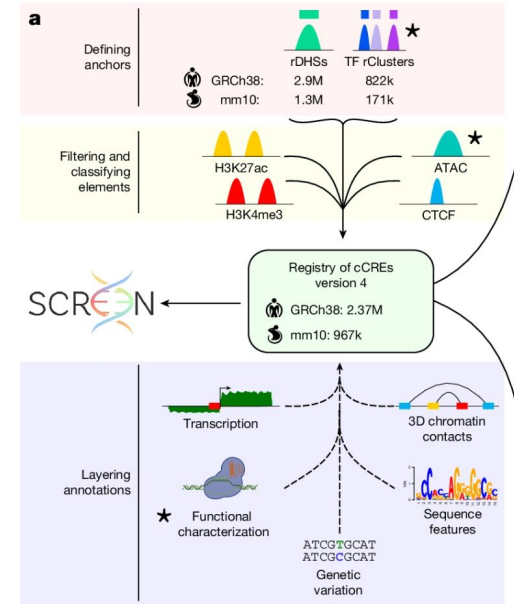
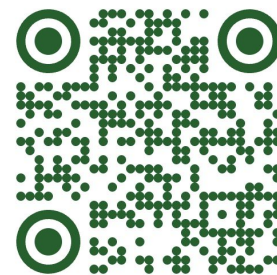
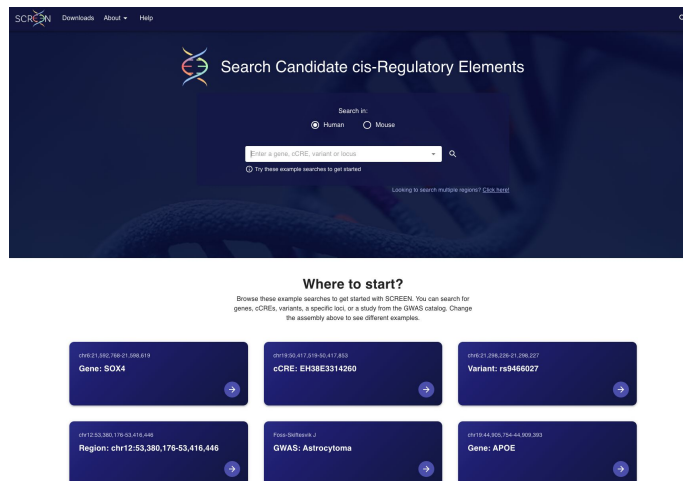


Figure extracted from *Expanded encyclopaedias of DNA elements in the human and mouse genomes*³

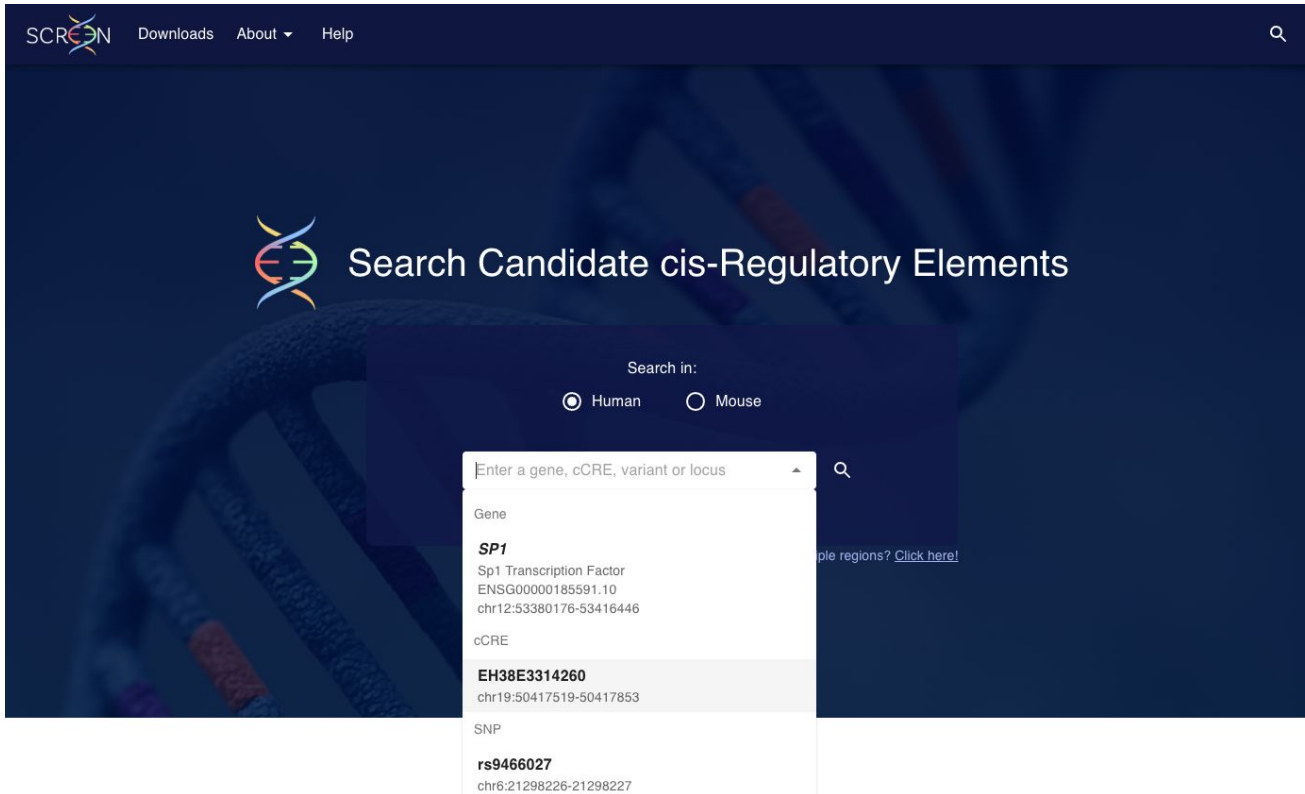
³Moore, J.E., Pratt, H.E., Fan, K. et al. An expanded registry of candidate cis-regulatory elements. *Nature* (2026). <https://doi.org/10.1038/s41586-025-09909-9>

SCREEN

- ✓ Web-based visualization and discovery platform for exploring the ENCODE Registry cCREs and other ENCODE Encyclopedia annotations.
- ✓ Enables users to query, visualize, and download regulatory annotations across hundreds of human and mouse cell and tissue types
- ✓ Turns the catalog into analysis: Examine cCREs relationships to genes, TF binding, chromatin state, and three-dimensional genome organization.



Accessing cCREs: SCREEN browser



The screenshot displays the SCREEN browser interface. At the top, there is a dark blue header with the SCREEN logo, navigation links for 'Downloads', 'About', and 'Help', and a search icon. The main content area has a dark blue background with a DNA double helix graphic. The title 'Search Candidate cis-Regulatory Elements' is prominently displayed. Below the title, a search box is shown with a dropdown menu. The search box contains the text 'Enter a gene, cCRE, variant or locus'. The dropdown menu is open, showing three categories: 'Gene', 'cCRE', and 'SNP'. Under 'Gene', the entry 'SP1' is highlighted, showing its full name 'Sp1 Transcription Factor', its Ensembl ID 'ENSG00000185591.10', and its genomic location 'chr12:53380176-53416446'. Under 'cCRE', the entry 'EH38E3314260' is highlighted, showing its genomic location 'chr19:50417519-50417853'. Under 'SNP', the entry 'rs9466027' is highlighted, showing its genomic location 'chr6:21298226-21298227'. A search icon is also present next to the search box.

SCREEN Downloads About Help

Search Candidate cis-Regulatory Elements

Search in:

☒ Human ☐ Mouse

Enter a gene, cCRE, variant or locus

Gene

SP1
Sp1 Transcription Factor
ENSG00000185591.10
chr12:53380176-53416446

cCRE

EH38E3314260
chr19:50417519-50417853

SNP

rs9466027
chr6:21298226-21298227

Accessing cCREs: SCREEN browser



SCREEN

Downloads About Help

Enter a gene, cCRE, variant or locus

EH38E3314260 +

UCSC

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cCRE Details

EH38E3314260

Proximal Enhancer | chr19:50,417,519-50,417,853

UCSC

Primary Classification

Additional Classification

DNase

ATAC

H3K4me3

H3K27ac

CTCF

Cell Type Agnostic Classification

DNase max-Z	ATAC max-Z	H3K4me3 max-Z	H3K27ac max-Z	CTCF max-Z	Classification
3.74	2.85	4.16	4.94	1.07	Proximal Enhancer

Classification Proportions, Core Collection:

Core Collection

Cell Type	Organ/Tissue	Sample Type	Life Stage	↓ DNase	ATAC	H3K4me3	H3K27ac	CTCF	TF	Classification	Assays
Pancreas, female child (16 years)	Pancreas	Tissue	adult	2.80	2.85	-0.50	0.96	0.02	--	CA	
OCH-LY7	Blood	Cell line	adult	2.71	--	3.11	3.31	-1.30	No	Proximal Enhancer	
Body of pancreas, male adult (37 years)	Pancreas	Tissue	adult	2.69	1.78	-0.24	1.92	-1.41	--	Proximal Enhancer	
Body of pancreas, female adult (51 years)	Pancreas	Tissue	adult	2.69	0.82	0.21	0.87	0.56	--	CA	
Peyer's patch, female adult (51 years)	Small intestine	Tissue	adult	2.53	--	2.82	4.05	0.48	--	Proximal Enhancer	
Pancreas, female adult (59 years)	Pancreas	Tissue	adult	2.43	2.32	0.81	0.51	-0.42	--	CA	
GM12878	Blood	Cell line	adult	2.21	0.80	2.86	2.41	0.46	Yes	Proximal Enhancer	

Total Rows: 170

Chromatin Accessibility, Partial Data Collection:

Partial Data Collection

Cell Type	Organ/Tissue	Sample Type	Life Stage	↓ DNase	ATAC	H3K4me3	H3K27ac	CTCF	TF	Classification	Assays
B cell, female adult (43 years)	Blood	Primary cell	adult	3.74	--	3.56	--	--	--	CA-H3K4me3	
NAMALWA, treated with Sendai virus for 2.0 hours	Blood	Cell line	adult	3.41	--	--	--	--	--	CA	
NAMALWA	Blood	Cell line	adult	3.37	--	--	--	--	--	CA	
B cell, male adult (21 years)	Blood	Primary cell	adult	3.26	--	--	--	--	--	CA	
Naive B cell	Blood	Primary cell	adult	3.19	--	2.32	2.47	--	--	Proximal Enhancer	
Pancreas, male adult (26 years)	Pancreas	Tissue	adult	3.03	--	--	--	--	--	CA	

Accessing cCREs: SCREEN browser



SCREEN Downloads About Help

EH38E3314260 +

cCRE Details

EH38E3314260
Proximal Enhancer l chr19:50,417,519-50,417,853

Primary Classification Additional Classification **DNase** ATAC H3K4me3 H3K27ac CTCF

Table View Bar Plot Violin Plot UMAP Download

EH38E3314260 DNase z-scores

Cell Type	DNase	Classification
☐ B cell, female adult (43 years)	3.74	CA-H3K4me3
☐ NAMALWA, treated with Sendai virus for 2.0 hours	3.41	CA
☐ NAMALWA	3.37	CA
☐ B cell, male adult (21 years)	3.26	CA
☐ Naive B cell	3.19	Proximal Enhancer
☐ Pancreas, male adult (26 years)	3.03	CA
☐ Stimulated activated naive B cell, female adult (39 year...	2.98	CA
☐ Right lobe of liver, female adult (47 years)	2.86	CA
☐ Small intestine, male adult (34 years)	2.82	Proximal Enhancer
☐ Pancreas, female child (16 years)	2.80	CA
☐ Naive B cell	2.80	Proximal Enhancer
☐ Memory B cell, male adult (40 years)	2.75	Proximal Enhancer
☐ GM06990	2.73	CA-H3K4me3
☐ OCH-LY7	2.71	Proximal Enhancer
☐ Body of pancreas, male adult (37 years)	2.69	Proximal Enhancer
☐ Body of pancreas, female adult (51 years)	2.69	CA
☐ Naive B cell, female adult (39 years)	2.65	Proximal Enhancer

Total Rows: 1,325

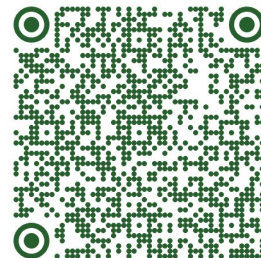
View By Value Tissue Tissue Max Hide Low Z-Scores On Off 95th Percentile Line (1.64) On Off

EH38E3314260 DNase z-scores

Low Signal 0 0.5 1 1.5 2 2.5 3 3.5

Blood B cell, female adult (43 years) NAMALWA, treated with ... NAMALWA B cell, male adult (21 years) Naive B cell Pancreas, male adult (26 years) Stimulated activated naive B cell, female adult (39 year...) Right lobe of liver, female adult (47 years) Small intestine, male adult (34 years) Pancreas, female child (16 years) Naive B cell Memory B cell, male adult (40 years) GM06990 OCH-LY7 Body of pancreas, male adult (37 years) Body of pancreas, female adult (51 years) Activated B cell, male adult (39 years) Naive B cell, with multi... Peyer's patch, female adult (39 years) Memory B cell, female adult (39 years) Ishikawa, treated with ... Pancreas, female adult (39 years) Left colon, female adult (39 years) B cell, female adult (39 years) Body of pancreas, female adult (39 years) CD4-positive, alpha-beta... Karpas-422 CD4-positive, alpha-beta... Pancreas, female adult (39 years) B cell, male adult (22 years) Colonic mucosa, female adult (39 years) Naive B cell, male adult (39 years) GM12878 Pancreas, female adult (39 years) Germinal center B cell, female adult (39 years)

Accessing cCREs: SCREEN browser



SCREEN

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Enter a gene, cCRE, variant or locus

EH38E3314260 +

cCRE

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cCRE Details

EH38E3314260

Proximal Enhancer | chr19:50,417,519-50,417,853

Closest Genes

Name	Type	Chromosome	Start	End	Distance to TSS
SPIB	Protein Coding	chr19	50418938	50431314	1,252
ENSG00000142539	Protein Coding	chr19	50415799	50428409	1,887
MYBPC2	Protein Coding	chr19	50432892	50466321	15,206

Intact Hi-C Loops

Common Gene Name	Gene Type	Experiment ID	Biosample	Score	$\uparrow$ P
NR1H2	Protein Coding	ENCSR529BKV	heart left ventricle,female adult (46 years)	55	0
NR1H2	Protein Coding	ENCSR181LJD	ovary,female adult (53 years)	28	0
NAPSA	Protein Coding	ENCSR395DZW	placenta,female embryo	35	0
ENSG00000269392	lncRNA	ENCSR395DZW	placenta,female embryo	35	0
NAPSA	Protein Coding	ENCSR898ANZ	adrenal gland,female adult (41 years)	32	0
ENSG00000269392	lncRNA	ENCSR898ANZ	adrenal gland,female adult (41 years)	32	0
NAPSA	Protein Coding	ENCSR692QWS	spleen,female adult (61 years)	57	0

Total Rows: 35

Accessing cCREs: SCREEN browser

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cCRE Details
EH38E3314260
Proximal Enhancer | chr19:50,417,519-50,417,853

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① No intersecting SNPs found in this region

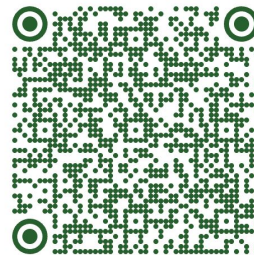
① This cCRE does not intersect any variants associated with significant changes in immune-related phenotypes

GTEX eQTLs for EH38E3314260

Variant Name	rsID	Gene	Chromosome	Position	Ref	Alt	Slope	Q Value ↑	Celltype
chr19_50417592_G_T_b38	rs3219450	ZNF473	chr19	50417592	G	T	0.65	5.96×10^{-4}	Skin Sun Exposed Lower leg
chr19_50417817_A_ACCCTGC_b38	rs3219456	KLK2	chr19	50417817	A	ACCCTGC	1.03	0.05	Spleen

Total Rows: 2

① No OneK1K eQTLs found



SCREEN Downloads About Help

EH38E3314260 +

cCRE Details
EH38E3314260
Proximal Enhancer | chr19:50,417,519-50,417,853

UCSC

Overview Conservation Simplex 240 Mammal Alignment

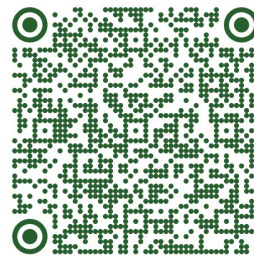
Genes
Variants
Conservation
Functional Characterization
TF Motifs

Conservation

	Vertebrates (100)	Mammals (241)	Primates (43)
PhyloP	0.52	2.10	0.44
PhastCons	0.32	0.57	0.60

Orthologous cCREs in mm10

Accession	Chromosome	Start	Stop
EM10E1646064	chr7	44532963	44533301



Accessing cCREs: SCREEN browser

EH38E3314260

+

cCRE

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EH38E3314260

Proximal Enhancer | chr19:50,417,519-50,417,853

Bound Transcription Factors

Overlapping ChIP-seq Peaks

TFs that bind to EH38E3314260

Transcription Factor

Total Experiments for TF

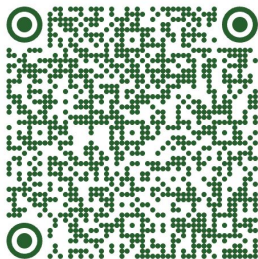
Bosample

ChIP-Seq Experiment Supporting Binding

IDR thresholded peaks

> POLR2AphosphoS5 (3)	45			
> EBF1 (2)	2			
> MAZ (2)	12			
> ZNF687 (2)	3			
> ATF7 (1)	5			
> BHLHE40 (1)	8			
> EP300 (1)	37			
> HDAC1 (1)	6			
> HNF4A (1)	5			
> IKZF2 (1)	2			
> KDM1A (1)	8			
> KLF1 (1)	2			
> KLF8 (1)	1			

Total Rows: 31



EH38E3314260

+

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EH38E3314260

Proximal Enhancer | chr19:50,417,519-50,417,853

Bound Transcription Factors

Overlapping ChIP-seq Peaks

Overlapping Representative ChIP-seq Peaks for transcription factors and other DNA-associated proteins

Transcription Factor

Motif Logo

Chromosome

Start

Stop

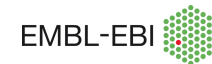
Motif Site Chromosome

Motif Site Start

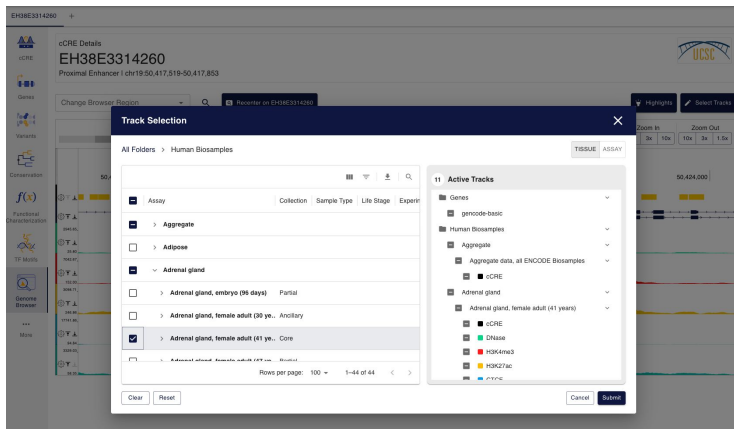
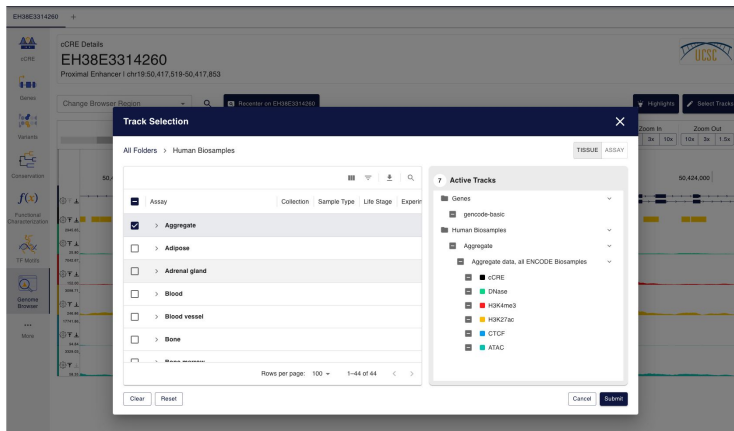
Motif Site Stop

ATF7 (2)		chr19	50,417,163	50,417,679	chr19	50,417,250	50,417,260
BHLHE40 (2)		chr19	50,417,385	50,417,789			
EBF1 (2)		chr19	50,417,519	50,417,617	chr19	50,417,568	50,417,579
EP300 (2)		chr19	50,417,492	50,417,732			
HDAC1 (2)		chr19	50,417,378	50,417,978			
HNF4A (2)		chr19	50,417,551	50,417,911	chr19	50,417,749	50,417,762

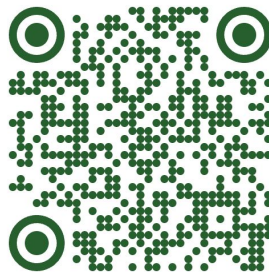
Total Rows: 31



Accessing cCREs: SCREEN browser



Accessing cCREs: SCREEN Downloads



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[Annotations](#) [Data Matrices](#) [Download cCREs in Genomic Region](#)

Human

cCREs by Class

cCREs by Cell and Tissue Type

cCRE-Gene Links

Functional Characterization

Mouse

Other

Human (GRCh38/hg38)
2,348,854 cCREs • 1,888 cell types

Download Human cCREs

Please cite our latest publication when using downloaded data: [Moore...Weng \(2026\), Nature](#)

cCREs by Class (.bed)

Promoter-like (PLS) (47,532)	2.6 MB
All Candidate Enhancers (pELS & dELS) (1,718,669)	94.4 MB
Proximal enhancer-like (pELS) (249,464)	13.7 MB
Distal enhancer-like (dELS) (1,469,205)	80.7 MB
Chromatin Accessible with CTCF (CA-CTCF) (126,034)	7.3 MB
Chromatin Accessible with H3K4me3 (CA-H3K4me3) (79,246)	4.8 MB
Chromatin Accessible with TF (CA-TF) (26,102)	1.5 MB
Chromatin Accessible Only (CA) (245,985)	13.0 MB
TF Only (TF) (105,286)	5.6 MB
CTCF-Bound cCREs (948,642)	63.0 MB
Silencer Sets (.tar.gz)	14.3 MB (unzipped)
MAFF/MAFK Dynamic Enhancers (.tar.gz)	827 KB (unzipped)

cCREs in Other Assemblies

GRCh37/hg19 (Lifted down from GRCh38/hg38)	128.8 MB
T2T CHM13v2.0/hs1 (Lifted over from GRCh38/hg38)	123 MB

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 ▾ Enter a gene, cCRE, variant or locus 🔍

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Functional Characterization

▼ Other

Orthologous cCREs

 **Human (GRCh38/hg38)**
2,348,854 cCREs • 1,888 cell types

Download Human cCREs ⬇

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cCREs by Cell and Tissue Type

Biosamples



Tissue/Biosample	Assays	Collection ↑	cCREs (.bed)	DNase Z-scores	ATAC Z-scores	H3K4me3 Z-scores	H3K27ac Z-scores	CTCF Z-scores
> Adipose (13)			⬇					
▼ Adrenal gland (19)			⬇					
Adrenal gland, female adult (41 years)		Core	⬇	⬇	⬇	⬇	⬇	⬇
Adrenal gland, female adult (51 years)		Core	⬇	⬇	⬇	⬇	⬇	⬇
Adrenal gland, female adult (53 years)		Core	⬇	⬇		⬇	⬇	⬇
Adrenal gland, male adult (37 years)		Core	⬇	⬇	⬇	⬇	⬇	⬇
Adrenal gland, male adult (54 years)		Core	⬇	⬇	⬇	⬇	⬇	⬇
Adrenal gland, embryo (96 days)		Partial	⬇	⬇				
Adrenal gland, female adult (47 years)		Partial	⬇	⬇	⬇			
Adrenal gland, female child (16 years)		Partial	⬇	⬇	⬇			
Adrenal gland, female embryo (108 days)		Partial	⬇	⬇				

Total Rows: 43

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Human

cCREs by Class

cCREs by Cell and Tissue Type

cCRE-Gene Links

Functional Characterization

Mouse

cCREs by Class

cCREs by Cell and Tissue Type

Functional Characterization

Other

Orthologous cCREs



Human (GRCh38/hg38)

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cCRE-Gene Associations

cCRE-Gene Links (3D Chromatin, CRISPR, eQTLs) (.zip)

463.3 MB 

cCRE-Gene-Links (ABC, EPIraction, GraphRegLR, rE2G) (.zip)

8.9 GB 

Nearest Gene by cCRE (protein-coding only) (.tsv)

317.5 MB 

Nearest Gene by cCRE (protein-coding and non protein-coding) (.tsv)

296.1 MB 



Downloads About Help



Enter a gene, cCRE, variant or locus



Annotations Data Matrices Download cCREs in Genomic Region

Human

cCREs by Class

cCREs by Cell and Tissue Type

cCRE-Gene Links

Functional Characterization

Mouse

cCREs by Class

cCREs by Cell and Tissue Type

Functional Characterization

Other

Orthologous cCREs



Human (GRCh38/hg38)

2,348,854 cCREs • 1,888 cell types

Download Human cCREs 

Please cite our latest publication when using downloaded data: [Moore...Weng \(2026\) Nature](#)

Functional Characterization

STARR seq Scores via CAPRA (whole genome)

1.79 GB (unzipped) 

MPRA

360.3 MB (unzipped) 

CRISPR Perturbation Data

35.7 MB (unzipped) 

VISTA Enhancers

123 KB 

Accessing cCREs: SCREEN Downloads

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 ▾ Enter a gene, cCRE, variant or locus ▾ 

Annotations Data Matrices Download cCREs in Genomic Region

Biosamples

Download

Filter

Search

Tissue/Biosample	Assays	Collection ↑
> Adipose (13)		
▼ Adrenal gland (19)		
Adrenal gland, female adult (41 years)		Core
Adrenal gland, female adult (51 years)		Core
Adrenal gland, female adult (53 years)		Core
Adrenal gland, male adult (37 years)		Core
Adrenal gland, male adult (54 years)		Core
Adrenal gland, embryo (96 days)		Partial
Adrenal gland, female adult (47 years)		Partial
Adrenal gland, female child (16 years)		Partial
Adrenal gland, female embryo (108 days)		Partial
Total Rows: 43		

Assembly

☒ GRCh38 ☐ mm10

Enter a genomic region

chr12:53380176-53416446

Selected Biosample: none

☒ Assays

☒ Conservation

☒ DNase

☒ Primate

☒ ATAC

☒ Mammal

☒ CTCF

☒ Vertebrate

☒ H3K27ac

☒ H3K4me3

Download Search Results (.bed) 

Hands-on



The ENCODE cCREs and SCREEN

- Explore the indicated locus on SCREEN:
 - GRCh38, chr8:127,682,026-127,744,101
 - GRCm39/mm39, chr6:6,740,753-6,891,103; GRCm38/mm10 chr6:6,740,753-6,891,103
- Inspect cCREs overlapping or near the indicated gene:
 - Human, MYC
 - Mouse, Dlx6
- Review cCRE class, biosample support, and linked evidence for:
 - Human, HCT116 core collection
 - Mouse, Forebrain E13.5 ancillary collection